

Study guide for Exam AI-901: Microsoft Azure AI Fundamentals

Purpose of this document

This study guide should help you understand what to expect on the exam and includes a summary of the topics the exam might cover and links to additional resources. The information and materials in this document should help you focus your studies as you prepare for the exam.

 Expand table

Useful links	Description
How to earn the certification	Some certifications only require passing one exam, while others require passing multiple exams.
Certification renewal	Microsoft associate, expert, and specialty certifications expire annually. You can renew by passing a free online assessment on Microsoft Learn.
Your Microsoft Learn profile	Connecting your certification profile to Microsoft Learn allows you to schedule and renew exams and share and print certificates.
Exam scoring and score reports	A score of 700 or greater is required to pass.
Exam sandbox	You can explore the exam environment by visiting our exam sandbox.
Request accommodations	If you use assistive devices, require extra time, or need modification to any part of the exam experience, you can request an accommodation.
Take a free Practice Assessment	Test your skills with practice questions to help you prepare for the exam.

Updates to the exam

Our exams are updated periodically to reflect skills that are required to perform a role. We have included two versions of the Skills Measured objectives depending on when you are taking the exam.

We always update the English language version of the exam first. Some exams are localized into other languages, and those are updated approximately eight weeks after the English version is updated. While Microsoft makes every effort to update localized exams as noted, there may be times when the localized versions of an exam are not updated on this schedule. Other available languages are listed in the **Schedule Exam** section of the **Exam Details** webpage. If the exam isn't available in your preferred language, you can request an additional 30 minutes to complete the exam.

Note

The bullets that follow each of the skills measured are intended to illustrate how we are assessing that skill. Related topics may be covered in the exam.

Note

Most questions cover features that are general availability (GA). The exam may contain questions on Preview features if those features are commonly used. You should be familiar with REST APIs, SDKs, and CLIs.

Skills measured as of April 15, 2026

Audience profile

As a candidate for this Microsoft Certification, you're at the beginning of your career in AI solution development.

For this exam, you should have conceptual knowledge of AI solutions in Azure and the foundational technical skills to work with them. You also need knowledge of Python coding syntax and programming techniques, and you should be familiar with Azure resources.

Skills at a glance

- Identify AI concepts and responsibilities (40–45%)
- Implement AI solutions by using Microsoft Foundry (55–60%)

Identify AI concepts and capabilities (40–45%)

Describe principles of responsible AI

- Describe considerations for fairness in an AI solution
- Describe considerations for reliability and safety in an AI solution
- Describe considerations for privacy and security in an AI solution
- Describe considerations for inclusiveness in an AI solution
- Describe considerations for transparency in an AI solution
- Describe considerations for accountability in an AI solution

Identify AI model components and configurations

- Describe how generative AI models work
- Identify an appropriate AI model, based on capabilities
- Identify appropriate model deployment options and configuration parameters

Identify AI workloads

- Identify scenarios for common AI workloads, including generative and agentic AI, text analysis, speech, computer vision, and information extraction

- Describe common text analysis techniques, including keyword extraction, entity detection, sentiment analysis, and summarization
- Identify features and capabilities of speech recognition and speech synthesis
- Identify features and capabilities of computer vision and image-generation models
- Identify techniques to extract information from text, images, audio, and videos

Implement AI solutions by using Microsoft Foundry (55–60%)

Implement generative AI apps and agents by using Foundry

- Create effective system and user prompts for generative AI models
- Deploy a model and interact with it in the Foundry portal
- Create a lightweight chat client application by using the Foundry SDK
- Create and test a single-agent solution in the Foundry portal
- Create a lightweight client application for an agent

Implement AI solutions for text and speech by using Foundry

- Build a lightweight application that includes text analysis
- Respond to spoken prompts by using a deployed multimodal model
- Build a lightweight application by using Azure Speech in Foundry Tools

Implement AI solutions with computer vision and image-generation capabilities by using Foundry

- Interpret visual input in prompts by using a deployed multimodal model
- Create new visual outputs by using generative models
- Build a lightweight application that includes vision capabilities

Implement AI solutions for information extraction by using Foundry

- Extract information from documents and forms by using Azure Content Understanding in Foundry Tools
- Extract information from images by using Content Understanding
- Extract information from audio and video by using Content Understanding
- Build a lightweight application with information extraction capabilities by using Content Understanding

Study resources

We recommend that you train and get hands-on experience before you take the exam. We offer self-study options and classroom training as well as links to documentation, community sites, and videos.

 Expand table

Study resources	Links to learning and documentation
Get trained	Choose from self-paced learning paths and modules or take an instructor-led course
Find documentation	Anomaly Detector Language Understanding Azure Machine Learning Computer Vision Natural language processing technology Azure Bot Service Speech to Text Speech Translation
Ask a question	Microsoft Q&A Microsoft Docs
Get community support	Artificial Intelligence and Machine Learning Hub
Follow Microsoft Learn	Microsoft Learn - Microsoft Tech Community
Find a video	The AI Show Browse other Microsoft Learn shows

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